

8. Many factors affect the rate of a chemical reaction.

(a) Explain why a change in concentration affects the rate of a reaction.

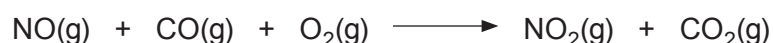
[2]

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(b) Under certain conditions nitrogen monoxide reacts with carbon monoxide and oxygen according to the equation below.



A study of the rate of this reaction, using varying initial concentrations of nitrogen monoxide and oxygen, gave the following data.

Experiment number	Concentration NO / mol dm ⁻³	Concentration O ₂ / mol dm ⁻³	Initial rate of reaction / mol dm ⁻³ s ⁻¹ × 10 ⁻⁴
1	1.0 × 10 ⁻⁴	1.0 × 10 ⁻⁴	4.40
2	2.0 × 10 ⁻⁴	1.0 × 10 ⁻⁴	17.6
3	3.0 × 10 ⁻⁴	1.0 × 10 ⁻⁴	39.6
4	2.0 × 10 ⁻⁴	2.0 × 10 ⁻⁴	17.6

(i) Use the data to determine how the concentration of NO affects the rate of the reaction. Explain your answer by referring to the experiment numbers.

[2]

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(ii) Use the data to determine how the concentration of O₂ affects the rate of the reaction. Explain your answer by referring to the experiment numbers.

[2]

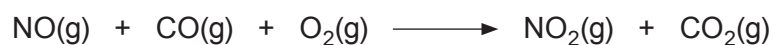
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- (iii) Suggest a method that could be used to follow changes over time as the reaction of nitrogen monoxide, carbon monoxide and oxygen proceeds. Explain your answer. [2]



- (iv) Describe the environmental implications on the atmosphere if the reaction above occurs on a large scale. [2]

- (v) The reaction shown is one of many that are catalysed in a catalytic converter. State where in a car a catalytic converter is used and name a suitable catalyst. [2]

Catalytic converter used in

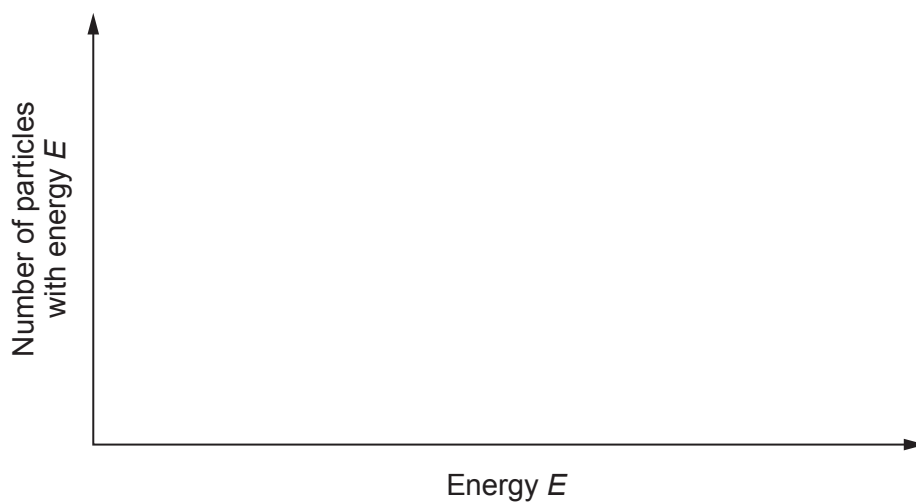
Catalyst used



- (c) On the axes below draw a Boltzmann distribution curve of the energies of molecules at a certain temperature.

Use this curve to explain how a catalyst increases the rate of a reaction.

[3]



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